Yanzheng Xiang

向彦铮

xyz1998seu@gmail.com
xyzcs.github.io

PROFESSIONAL SUMMARY

My current research focuses on Auto Research, spanning reliable code reproduction from academic papers, the development of LLM-based agents that propose novel scientific ideas and predict promising research directions, and the use of reinforcement learning to enhance these capabilities. I am also a direct contributor to Meta's Muse Glimmer and Muse Spark models.

EXPERIENCE

Meta | Research Engineer Nov. 2025–Sep. 2026

- Core contributor to Muse Glimmer, an open agentic foundation model (technical report).
- Contributed to the development of Muse Spark.
- Developing AI Scientist agents: LLM-based systems that automate scientific research tasks and machine learning engineering workflows.
- Building automated bug-injection pipelines for SWE-bench to construct training and evaluation datasets.
- Building an automated data construction pipeline for recursive self-improvement (RSI) that identifies model-specific weaknesses and generates targeted training data to improve model capabilities.

AstraZeneca | Internship Jul. 2025–Oct. 2025

Supervised by Dr. Saseendran and Dr. Jin. Conducted research on efficient inference for diffusion language models. Published COVER at ICML 2026, achieving up to 11.64× decoding speedup while preserving generation quality.

EDUCATION

King's College London Sep 2023–Sep 2027 (Expected)

Ph.D. in Computer Science
Supervised by Prof. Yulan He with Dr. Lin Gui as Co-advisor.
Topic: Large Language Model, In-context Learning, Interpretability.

Southeast University Sep 2020–Jun 2023

M.S. in Software Engineering
Average Score: 86.42 / 100
Supervised by Prof. Deyu Zhou.
Topic: Question Answering, Code Generation.

Hefei University of Technology Sep 2016–Jun 2020

B.E. in Computer Science and Technology

PUBLICATIONS

Technical Report 2026 **Muse Glimmer 30B: An On-Device Agentic Model.**
Muse Glimmer team (including **Yanzheng Xiang**, core contributor).

ICML 2026 **Stop the Flip-Flop: Context-Preserving Verification for Fast Revocable Diffusion Decoding.**
Yanzheng Xiang, Lan Wei, Yizhen Yao, Qinglin Zhu, Hanqi Yan, Chen Jin, Philip Alexander Teare, Dandan Zhang, Lin Gui, Amrutha Saseendran, Yulan He.

COLM 2025 **SciReplicate-Bench: Benchmarking LLMs in Agent-driven Algorithmic Reproduction from Research Papers.**
Yanzheng Xiang, Hanqi Yan, Shuyin Ouyang, Lin Gui, Yulan He.

Findings of ACL'23 **G3R: A Graph-Guided Generate-and-Rerank Framework for Cross-domain Text-to-SQL Generation.**
Yanzheng Xiang, Qian-Wen Zhang, Xu Zhang, Zejie Liu, Yunbo Cao, Deyu Zhou.

PUBLICATIONS (CONTINUED)

Findings of ACL'24 **Addressing Order Sensitivity of In-Context Demonstration Examples in Causal Language Models.**
Yanzheng Xiang, Hanqi Yan, Lin Gui, Yulan He.

NeurIPS 2026 **Latent Refinement Decoding: Enhancing Diffusion-Based Language Models by Refining Belief States.**
Qinglin Zhu, Yizhen Yao, Runcong Zhao, **Yanzheng Xiang**, Amrutha Saseendran, Chen Jin, Philip Teare, Bin Liang, Yulan He, Lin Gui.

EMNLP 2026 Oral **Follow the Latent Roadmap: Navigating Revocable Decoding for Diffusion LLMs with Anchor Tokens.**
Yizhen Yao, Qinglin Zhu, Runcong Zhao, Xiangxiang Dai, **Yanzheng Xiang**, Yulan He, Lin Gui.

EMNLP'24 **Encourage or Inhibit Monosemanticity? Revisit Monosemanticity from a Feature Decorrelation Perspective.**
Hanqi Yan, **Yanzheng Xiang**, Guangyi Chen, Yifei Wang, Lin Gui, Yulan He.

EMNLP'24 **The Mystery of In-Context Learning: A Comprehensive Survey on Interpretation and Analysis.**
Yuxiang Zhou, Jiazheng Li, **Yanzheng Xiang**, Hanqi Yan, Lin Gui, Yulan He.

Findings of ACL'25 **PECAN: LLM-Guided Dynamic Progress Control with Attention-Guided Hierarchical Weighted Graph for Long-Document QA.**
Xinyu Wang, **Yanzheng Xiang**, Lin Gui, Yulan He.

Findings of EMNLP'21 **A Divide-And-Conquer Approach for Multi-label Multi-hop Relation Detection in Knowledge Base QA.**
Deyu Zhou, **Yanzheng Xiang**, Linhai Zhang, Chenchen Ye, Qian-Wen Zhang, Yunbo Cao.

AWARDS AND HONORS

NMES International Studentship 2023–2027
Outstanding Graduate Student, Hefei University of Technology 2020
Merit Student, Hefei University of Technology 2018

INVITED TALKS

LLaMA Community Meet-up Apr. 6, 2025
Invited talk at Meta: "Towards Automatic Code Reproduction for Scientific Papers: Benchmarks and Methodologies."

COMPETITIONS

National First Prize (Top 0.65%), China Undergraduate Mathematical Contest in Modelling 2018
The contest challenges teams of up to three students to analyze, model, solve, and present solution reports to an open-ended application problem.

1st Place, Spider Leaderboard 2022
Spider is a large-scale semantic parsing dataset. Our model G3R achieved 1st place on the "exact set match without values"metric. Currently ranked 5th.

COURSE TEACHING

Discrete Mathematics, Southeast University 2021
Computer Programming for Data Scientists, King's College London 2024

OTHER SKILLS

IT: C++, SQL, Python, LaTeX
Languages: English (Fluent, IELTS 7.0), Chinese (Native)